A Revised Parsimonious Bayesian Model of Terrorist Targeting and State Escalation

Target-Specific Attack Frequency, Regime Type, Belief Updating, and the Police-to-Military Threshold

Table of contents

1 Theoretical Problem 2

2 Players and Regime Types 3

3 Incomplete Information About Terrorist Capacity 3

4 Terrorist Actions 4

5 Bayesian Updating 5

6 Security Targeting and Capacity Inference 5

7 State Response 6

8 Common Security Calculus 6

9 Regime-Specific Political Incentives 7
- 9.1 Democracies . . . 7
- 9.2 Autocracies . . . 7

10 State Escalation Condition 7

11 Posterior Escalation Threshold 8

12 Threshold Regions 8
- 12.1 Low-Threat Region . . . 8
- 12.2 Intermediate-Threat Region . . . 9

12.3 High-Threat Region 9

13 Basic Regime-Target Crossover **9**

14 Revised Crossover With Bayesian Updating **10**

15 Autocratic Dual Mechanism **10**

16 Target-Specific Frequency Thresholds **11**

17 Terrorist Payoff **11**

18 Equilibrium Interpretation **12**

19 Hypotheses **12**
19.1 H1: Democratic Civilian-Target Frequency Hypothesis 12
19.2 H2: Autocratic Security-Target Frequency Hypothesis 12
19.3 H3: Regime-Target Crossover Hypothesis 12
19.4 H4: Capacity-Inference Hypothesis 13
19.5 H5: Autocratic Dual-Mechanism Hypothesis 13
19.6 H6: Differential Escalation-Threshold Hypothesis 13
19.7 H7: Bayesian-Shift Hypothesis 13
19.8 H8: Threshold-Contingency Hypothesis 13
19.9 H9: Selective State-Capacity Hypothesis 13

20 Empirical Representation **14**

21 Conclusion **14**

1 Theoretical Problem

States do not respond uniformly to terrorism. Some terrorist campaigns remain primarily matters for police, intelligence agencies, and courts, while others lead governments to mobilize or deploy military forces.

The central argument developed here is that terrorist targeting affects state responses through two mechanisms.

First, targets have different **political meanings across regime types**:

- In democracies, attacks against civilians can generate public-accountability pressures.
- In autocracies, attacks against military, police, and security institutions can generate regime-survival anxiety.

Second, target choice may provide governments with **information about terrorist organizational capacity**. The terrorist organization knows its own strength, but the government observes that strength imperfectly. Governments therefore update their beliefs from observable terrorist behavior.

The theoretical sequence is:

$$\text{Target-specific attack frequency} \rightarrow \begin{cases} \text{political pressure,} \\ \text{belief updating about terrorist capacity} \end{cases} \rightarrow \text{state response.}$$

The state's ultimate choice is whether to continue treating terrorism primarily as a police/law-enforcement problem or escalate toward a military response.

2 Players and Regime Types

There are two strategic actors:

1. a terrorist organization, T;
2. a state government, G.

The state has an exogenously given regime type

$$r \in \{D, A\},$$

where D denotes democracy and A denotes autocracy. Regime type is common knowledge.

The model does not assume that democracies or autocracies are inherently more aggressive. Instead, regime type changes the political consequences associated with different terrorist targets.

3 Incomplete Information About Terrorist Capacity

Nature determines terrorist organizational capacity:

$$\theta \in \{L, H\},$$

where L denotes a low-capacity organization and H denotes a high-capacity organization.

The prior probability of a high-capacity organization is

$$\Pr(\theta = H) = \pi,$$

with

$$\Pr(\theta = L) = 1 - \pi.$$

The terrorist organization observes its own type. The state does not.

Let

$$q_H > q_L$$

represent the expected security threat associated with high- and low-capacity organizations.

4 Terrorist Actions

The terrorist organization chooses a campaign along two dimensions:

1. target type;
2. attack frequency.

Let

$$x \in \{C, S\},$$

where C denotes civilian targets and S denotes military/security-force targets.

The organization also chooses

$$f \in \{0, 1\},$$

where $f = 0$ represents status-quo frequency and $f = 1$ represents increased attack frequency.

The terrorist action is therefore

$$a = (x, f).$$

The four observable campaign profiles are:

Action	Interpretation
$C0$	Civilian targeting at status-quo frequency
$C1$	Increased frequency of civilian attacks
$S0$	Security-force targeting at status-quo frequency
$S1$	Increased frequency of security-force attacks

This formulation makes target-specific attack frequencies the theoretically relevant empirical quantities.

5 Bayesian Updating

The state observes $a = (x, f)$ but does not directly observe terrorist capacity. It forms the posterior belief

$$\mu(a) = \Pr(H \mid a).$$

For an equilibrium-path action,

$$\mu(a) = \frac{\pi\sigma_H(a)}{\pi\sigma_H(a) + (1-\pi)\sigma_L(a)}.$$

The model does not assume that terrorists deliberately choose targets to signal strength. Instead, information can emerge because different terrorist types choose different actions with different probabilities.

6 Security Targeting and Capacity Inference

Suppose high-capacity organizations are relatively more capable of attacking security institutions than low-capacity organizations. Then

$$\Pr(S \mid H) > \Pr(S \mid L).$$

Bayes' rule implies

$$\Pr(H \mid S) > \Pr(H).$$

Thus,

$$\mu_S > \pi.$$

Rarity alone, however, does not make security targeting informative. The relevant quantity is the likelihood ratio

$$\frac{\Pr(S \mid H)}{\Pr(S \mid L)}.$$

Security targeting provides information about terrorist capacity only when it is relatively more characteristic of high-capacity organizations.

7 State Response

After observing the terrorist campaign and updating its beliefs, the state chooses

$$y \in \{P, M\},$$

where P denotes a limited or police-centered response and M denotes military escalation.

8 Common Security Calculus

Expected terrorist threat is

$$Q(\mu) = q_L + (q_H - q_L)\mu(a).$$

Let $\phi > 0$ represent the additional security pressure generated by increased attack frequency. Let τ represent a regime-neutral difference between civilian and security targeting. Let $c(k)$ represent the relative cost of military escalation compared with policing, conditional on known state capacity k.

The common security component is therefore

$$Q(\mu) + \phi f + \tau \mathbf{1}\{x = C\} - c(k).$$

9 Regime-Specific Political Incentives

9.1 Democracies

For democracies,

$$\Pi_D(x, f) = d_C \mathbf{1}\{x = C\} + d_{CF} \mathbf{1}\{x = C\} f.$$

The parameter $d_C > 0$ captures public-accountability pressure generated by civilian victimization. The parameter $d_{CF} \geq 0$ captures the possibility that this pressure increases as civilian attacks become more frequent.

9.2 Autocracies

For autocracies,

$$\Pi_A(x, f) = a_S \mathbf{1}\{x = S\} + a_{SF} \mathbf{1}\{x = S\} f.$$

The parameter $a_S > 0$ captures regime-survival anxiety generated by attacks against the coercive apparatus. The parameter $a_{SF} \geq 0$ captures the possibility that repeated attacks against security institutions increasingly threaten perceptions of regime control.

10 State Escalation Condition

The state's net payoff advantage from military escalation is

$$\Delta_r(a, \mu; k) = Q(\mu) + \phi f + \tau \mathbf{1}\{x = C\} + \Pi_r(x, f) - c(k).$$

The state chooses military escalation whenever

$$\Delta_r(a, \mu; k) \geq 0.$$

Otherwise, it chooses the police-centered response.

11 Posterior Escalation Threshold

Substituting the expected-threat function gives

$$q_L + (q_H - q_L)\mu(a) + \phi f + \tau \mathbf{1}\{x = C\} + \Pi_r(x, f) - c(k) \geq 0.$$

Solving for the posterior belief gives

$$\mu(a) \geq \mu_r^*(x, f; k),$$

where

$$\mu_r^*(x, f; k) = \frac{c(k) - q_L - \phi f - \tau \mathbf{1}\{x = C\} - \Pi_r(x, f)}{q_H - q_L}.$$

If

$$\mu(a) < \mu_r^*,$$

the state treats terrorism primarily as a police problem. If

$$\mu(a) \geq \mu_r^*,$$

the state escalates militarily.

12 Threshold Regions

12.1 Low-Threat Region

When perceived terrorist capacity lies below both target-specific thresholds, the state chooses policing regardless of target type.

12.2 Intermediate-Threat Region

For a democracy, suppose

$$\mu_D^*(C, f) < \mu < \mu_D^*(S, f).$$

Then civilian targeting produces military escalation while security targeting remains a police problem.

For an autocracy, the ordering may reverse:

$$\mu_A^*(S, f) < \mu < \mu_A^*(C, f).$$

Then security targeting produces military escalation while civilian targeting remains a police problem.

12.3 High-Threat Region

When perceived terrorist capacity exceeds both thresholds, both target types produce military escalation. Target type should therefore have its strongest observable effect near the escalation threshold.

13 Basic Regime-Target Crossover

Holding posterior beliefs constant, democracies prefer escalation relatively more after civilian attacks when

$$\tau + d_C + d_{CF} f > 0.$$

Autocracies prefer escalation relatively more after security attacks when

$$\tau - a_S - a_{SF} f < 0.$$

Both conditions hold when

$$-(d_C + d_{CF} f) < \tau < a_S + a_{SF} f.$$

14 Revised Crossover With Bayesian Updating

Let

$$\mu_C = \Pr(H \mid C)$$

and

$$\mu_S = \Pr(H \mid S).$$

If security targeting is relatively more characteristic of strong organizations,

$$\mu_S > \mu_C.$$

Define

$$B = (q_H - q_L)(\mu_S - \mu_C).$$

The generalized crossover condition becomes

$$\boxed{B - (d_C + d_{CF} f) < \tau < B + (a_S + a_{SF} f)}.$$

This is the central theoretical result. Target type affects state escalation through two channels:

$$\text{Target type} \rightarrow \begin{cases} \text{regime-specific political pressure,} \\ \text{information about terrorist capacity.} \end{cases}$$

15 Autocratic Dual Mechanism

For autocracies, security targeting can simultaneously increase regime-survival anxiety and perceived terrorist capacity:

$$S \rightarrow \begin{cases} a_S \uparrow, \\ \mu_S \uparrow \end{cases} \rightarrow M.$$

The two mechanisms reinforce one another.

16 Target-Specific Frequency Thresholds

Let

$$f^*_{xr}$$

represent the target-specific frequency threshold at which regime r switches from policing to military escalation.

Under the relevant parameter restrictions, the model predicts for autocracies:

$$\boxed{f^*_{SA} < f^*_{CA}}.$$

Equivalently, the difference in autocratic escalation incentives between security and civilian targeting increases with

$$(q_H - q_L)(\mu_S - \mu_C) - \tau.$$

17 Terrorist Payoff

Let terrorist payoff be

$$U^\theta_T(a, y) = G_\theta(a) - D_\theta(a)\mathbf{1}\{y = M\}.$$

There is no direct payoff from changing the state's belief. Beliefs matter indirectly:

$$a \to \mu(a) \to y \to U_T.$$

The terrorist type chooses

$$a^*_\theta \in \arg\max_a \left[G_\theta(a) - D_\theta(a)m_r(a)\right],$$

where

$$m_r(a) = \mathbf{1}\{\mu(a) \geq \mu^*_r(a; k)\}.$$

18 Equilibrium Interpretation

The model can generate pooling, separating, or semi-separating equilibria. In a pooling equilibrium,

$$\mu(a) = \pi.$$

In a separating equilibrium, observed behavior can reveal terrorist capacity. Semi-separating equilibria generate partial Bayesian updating. The empirical theory requires only that some campaign behaviors be relatively more likely among high-capacity organizations.

19 Hypotheses

19.1 H1: Democratic Civilian-Target Frequency Hypothesis

As the frequency of civilian-targeted terrorist attacks increases, the probability of military escalation should increase more strongly in democracies than in autocracies.

$$\left.\frac{\partial \Pr(M)}{\partial F_C}\right|_D > \left.\frac{\partial \Pr(M)}{\partial F_C}\right|_A.$$

19.2 H2: Autocratic Security-Target Frequency Hypothesis

As the frequency of attacks against military and security forces increases, the probability of military escalation should increase more strongly in autocracies than in democracies.

$$\left.\frac{\partial \Pr(M)}{\partial F_S}\right|_A > \left.\frac{\partial \Pr(M)}{\partial F_S}\right|_D.$$

19.3 H3: Regime-Target Crossover Hypothesis

The relationship between target-specific terrorist attack frequency and military escalation should reverse across regime types: civilian-targeted attack frequency should generate stronger escalation incentives in democracies, whereas security-force-targeted attack frequency should generate stronger escalation incentives in autocracies.

19.4 H4: Capacity-Inference Hypothesis

Security-force targeting should increase the probability of military escalation when such attacks cause governments to infer that the terrorist organization possesses greater organizational capacity. The mechanism requires

$$\Pr(S \mid H) > \Pr(S \mid L).$$

19.5 H5: Autocratic Dual-Mechanism Hypothesis

The effect of security-force targeting on military escalation should be especially strong in autocracies because attacks against coercive institutions simultaneously increase regime-survival anxiety and may increase perceived terrorist capacity.

19.6 H6: Differential Escalation-Threshold Hypothesis

In autocracies, the frequency of security-force attacks required to trigger military escalation should be lower than the frequency of civilian attacks required to trigger military escalation:

$$f^*_{SA} < f^*_{CA}.$$

19.7 H7: Bayesian-Shift Hypothesis

The autocratic security-target advantage should increase as the inferred capacity difference between security and civilian targeting grows:

$$(q_H - q_L)(\mu_S - \mu_C) - \tau > 0.$$

19.8 H8: Threshold-Contingency Hypothesis

The effects of target type and regime type should be strongest when the state is near its police-to-military escalation threshold and weaker when perceived terrorist capacity is either very low or very high.

19.9 H9: Selective State-Capacity Hypothesis

Greater police, intelligence, bureaucratic, and judicial capacity should increase the level of terrorist threat necessary to induce military escalation.

20 Empirical Representation

Define

$$F_C = \text{frequency of civilian-targeted attacks}$$

and

$$F_S = \text{frequency of security-force-targeted attacks.}$$

A reduced-form empirical model is

$$\Pr(M_{it} = 1) = F\big(\alpha + \beta_1 F_{C,it} + \beta_2 F_{S,it} + \beta_3 D_i + \beta_4 F_{C,it} D_i + \beta_5 F_{S,it} D_i + \beta_6 \hat{\mu}_{it} + \beta_7 k_{it} + \mathbf{X}_{it}\gamma\big)\,.$$

The theoretical predictions concern marginal effects. For civilian attacks:

$$\left.\frac{\partial \Pr(M)}{\partial F_C}\right|_D > \left.\frac{\partial \Pr(M)}{\partial F_C}\right|_A .$$

For security attacks:

$$\left.\frac{\partial \Pr(M)}{\partial F_S}\right|_A > \left.\frac{\partial \Pr(M)}{\partial F_S}\right|_D .$$

21 Conclusion

The model proposes that target-specific terrorist attack frequency affects state escalation through three linked processes:

$$\boxed{\text{Target-specific attack frequency}}$$

affects

$$\boxed{\text{regime-specific political vulnerability}}$$

and

government beliefs about terrorist capacity,

which jointly determine whether the state crosses

the police-to-military escalation threshold.

The most distinctive implication is that the same observable terrorist campaign can have different political meanings across regimes while simultaneously providing governments with information about terrorist organizational capacity.

For autocracies in particular,

security targeting $\rightarrow$ survivability anxiety + higher perceived terrorist capacity $\rightarrow$ military escalation.

This provides a theoretical explanation for why an autocratic government could tolerate a comparatively substantial civilian terrorist campaign as a police problem while responding militarily to a much smaller number of attacks against institutions central to regime security.